

Numerical Methods in Chemical Engineering and Problem Solving

ChE 348 (Unique # 16500)

Spring 2026

Course Description: The purpose of this course is to learn basic computational methods for solving a variety of mathematical problems that cannot be solved analytically, and to develop your confidence in using computational tools for problem solving in engineering. The methods will be applied to several problems in chemical engineering.

Prerequisites: CHE 210, CHE 317, and M 427K.

Lectures: T Th 12:30PM–2:00PM (CPE 2.218)

Recitations: F 11:00AM–12:00PM (ETC 2.136)

Instructor: Dr. Nathaniel A. Lynd lynd@che.utexas.edu
Office Hours: TBD
Location: GLT 1.234 Office Phone: 512-232-5238

Teaching Assistant: Tyler Bailey tjb4283@my.utexas.edu
Office Hours: TBD
GLT

Course website: on *Canvas* <http://canvas.utexas.edu>

Lecture software: Matlab, Mathematica

Required Text: *None.* Excerpts will be given out for reading. The files will be uploaded to Canvas, and are continually revised. Check back for updates!

Required Computer: Please see the departmental requirement for student computers [\[link\]](#).
A Chromebook or iPad will not be helpful.

Assessment:

Quizzes	30 % (top six)
Midterm #1	20 %
Midterm #2	15 %
Midterm #3	15 %
Last Exam	20 %

Anticipated course grade cut-offs are:

A	> 94%
A–	> 90%
B+	> 87%
B	> 84%
B–	> 80%
C+	> 77%
C	> 74%
C–	> 70%

These grade cut-offs may be adjusted by the instructor.

Homework is not collected, but is highly recommended that you do it. Problem sets will be worked in a collaborative fashion with the TA during recitation. Selected homework problems will also be worked during lecture. The balance you should do on your own, as aspects of those problems will show up on quizzes and midterms.

Quizzes:

Quizzes will be given both in lecture and as take-home assignments approximately weekly to fortnightly. The top six quiz scores will be counted toward your final grade. No make-up quizzes will be given for any reason and it is expected that only counting the top six quizzes (out of up to 14 that will be assigned) will allow for occasional absence due to illness or other UT sponsored activities.

Exam Dates: Code Exam (Midterm #1): [February 3rd](#), in lecture
Midterm #2: [March 10th](#), in lecture
Midterm #3: [April 23rd](#), in lecture
Final Exam: [May 4th](#), 1:00PM–3:00PM

Notify instructor by January 28th, 2026 if you have a time conflict during any of these exam periods.

Last Exam Policy: If you are earning above a 90% by the time of the Last Exam, no last examination is required.

Make-up exams will generally never be given for any reason. In some instances, a make-up exam will be given in extenuating circumstances *documented by a letter from UT Austin*. Notify instructor as soon as possible *before the midterm*.

COURSE EXPECTATIONS

Knowledge, abilities and skills students should have entering this course:

1. **This course has the following strict prerequisites:** CHE 210, CHE 317 and M 427K with a grade of C at least in each of them. The **only** exceptions to this policy are students who have already acquired permission from me to waive the prerequisites.
2. Complete understanding of **algebra, trigonometric and logarithmic functions** (Math 305G)
3. Familiarity with **matrices** and their properties (CHE 210).
4. Ability to work with and manipulate **partial derivatives, integration, Taylor series expansions** (Math 408C, 408D, 427).
5. Familiarity with methods to analytically solve linear **ordinary differential equations of first and second order** (Math 408C, 408D, 427).
6. Understanding of **scientific units, absolute temperature** (CHE 317)
7. Complete understanding of the **behavior of ideal gases** and some familiarity with the **properties of real gases** (CHE 317, CH 302).
8. Fundamental understanding of **mass and energy balances** (CHE 317)
9. **Stoichiometry and equilibrium constants** (Ch 301, 302, 353)

Knowledge, abilities and skills students should gain from this course:

This course has several objectives. The foremost is to introduce several computational techniques which are important in the solution of a variety of mathematical problems that cannot be solved analytically. By the end of this course, students should be able to:

1. Successfully solve numerical engineering problems with a computer.
2. Be able to formulate and write structured code in MATLAB.
3. Understand the foundation behind the basic numerical methods for matrix manipulations.
4. Solve sets of linear and nonlinear equations using numerical methods as well as built-in MATLAB functions.
5. Apply numerical methods and MATLAB functions to differentiate and integrate a function or a set of discrete points.
6. Apply explicit and implicit numerical methods and MATLAB functions to integrate single and multiple sets of initial value problems.
7. Apply shooting and finite difference methods to solve boundary value problems.
8. Apply finite difference and implicit methods to solve parabolic and elliptic partial differential equations.
9. Apply numerical methods and MATLAB functions to perform rudimentary optimization in the context of parameter estimation.
10. Ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
 - A complex engineering problem is: Involving wide-ranging or conflicting technical issues, having no obvious solution, addressing problems not encompassed by current standards and codes, involving diverse groups of stakeholders, including many component parts or sub-problems, involving multiple disciplines, or having significant consequences in a range of contexts.
11. Ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
 - The intent is to include all of these issues in the same design – however, it is not necessary that they are all relevant.

The programming environment incorporated into MATLAB will be used to introduce structured programming techniques. While you will be required to do your work in MATLAB for this course, the major concepts taught in this course are broadly transferable to other software.

Software: You must download MATLAB onto your own computer. It is available free-of-charge to UT students at:

<https://www.utexas.edu/its/products/matlab/>

You may need ZOOM in order to watch lecture recordings or to virtually participate in lecture if the instructor falls ill.

It is highly suggested that you also download Mathematica onto your own laptop. It is available free-of-charge to UT students at:

<https://cloud.wikis.utexas.edu/wiki/spaces/eceit/pages/37755100/Wolfram+s+Mathematica>

TENTATIVE COURSE SCHEDULE

Date	Topic	Notes
Topic #1: Programming		
01/13	Course overview, MATLAB	
01/15	MATLAB and programming	
Topic #2: Linear systems of equations		
01/20	Introduction to systems of linear equations and applications	
01/22	Introduction to systems of linear equations and applications	
01/27	Expo – no lecture	
01/29	Matrix properties and algebra; Gaussian Elimination	
02/03	Midterm #1: Matlab exam	
02/05	LU decomposition, Eigenvalues, Tridiagonal Matrices Application: Liquid-liquid extractor	
Topic #3: Non-linear systems of equations		
02/10	Successive Substitution, Newton and Secant Methods Review of Taylor series expansion	
02/12	Bounded methods: Bisection, False position method	
02/17	Case Study: Phase Equilibrium (Computation of a vapor-liquid envelope)	
02/19	Generalized Newton-Raphson Method Application: Multiple reactions in a CSTR	
Topic #4: Interpolation & Integration		
02/24	Interpolation and Brent's methods Application: Vapor pressure data representation	
02/26	Numerical integration and interpolation	
03/05	Numerical integration. Applications: Distillation, and two electrons in a box	
03/10	Midterm #2	
Topic #5: Ordinary differential equations (Initial value problems)		
03/12	Explicit and implicit Euler Method and numerical stability , Higher order and predictor – corrector methods, stiffness, Runge-Kutta methods	
03/17	Spring Break	

03/19	Spring Break
03/24	Applications: Multiple reactions in a batch reactor and in a plug flow reactor
Topic #6: Ordinary differential equations (Boundary value problems)	
03/24	Introduction to BVPs, Finite difference method
03/31	Application: Reaction/diffusion in a slab
04/02	Shooting method
04/07	Application: reaction/diffusion in a catalytic pellet. Incorporating derivative boundary conditions
Topic #7: Partial differential equations (Parabolic + Elliptical problems)	
04/09	Explicit Central Finite Difference Method Application: Reaction/diffusion in a membrane (Parabolic PDEs) Stability and Implicit Methods (Parabolic PDEs)
04/14	Application: Heat conduction in a rectangular plate (Elliptic PDEs), relaxation methods
04/17	ODEs and PDEs: Review and additional applications. Case study: My experience in modeling polymer self-assembly: Numerical steps and missteps
04/21	Case study: My experience in modeling polymer self-assembly: Numerical steps and missteps
Topic #8: Optimization	
04/21	Optimization
04/23	Midterm #3
05/04	Final Exam, 1:00–3:00PM

FEEDBACK

It is my goal to make this an excellent course and a positive experience for all of us. If you feel that the course is not meeting your expectations or you want to provide feedback on how the course is progressing for you, I encourage you to contact me as early as possible.

POLICIES

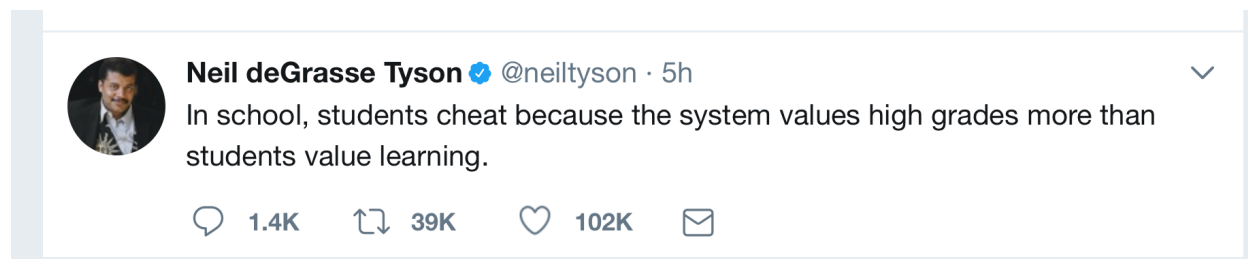
Recitations: The teaching assistant or instructor will conduct recitations, where MATLAB syntax and coding will be reviewed and/or a problem similar to problems on the homework will be given to be solved. This is an excellent opportunity to get advice while you are working on a computer. Please go to recitation and take advantage of it.

Participation/Professionalism: I expect students to show up on time for lecture, to be courteous to their neighbors (and their instructor) and only discuss material while the class is discussing a

topic. It is extremely distracting to show up late, particularly during a quiz. Cell phones should not be used during lecture. Watching YouTube videos or Netflix during lecture is distracting to those behind you.

Electronics: Set your cell phone to silent, and put it in your bag. **You are encouraged to bring your laptop to lecture.** However, its primary use should be to follow along with programming exercises and not to be a distraction to those around you.

Academic Integrity



University of Texas Honor Code: The core values of The University of Texas at Austin are learning, discovery, freedom, leadership, individual opportunity, and responsibility. Each member of the university is expected to uphold these values through integrity, honesty, trust, fairness, and respect toward peers and community.

Cheating encompasses such behaviors as copying another student's work, unauthorized use of large language models (i.e., "AI"), bringing materials into an exam that are not allowed, and a variety of other academically dishonest behaviors (a list of examples is included below). As with plagiarism, students who are discovered to have cheated will receive no credit for an assignment or exam and will be reported to Student Conduct and Academic Integrity without hesitation or prior warning. The referral will also be accompanied by a sanction that may consist of a one-letter grade reduction in course grade or more. The referral can also be as severe as a grade of F in the class. Once accused of academic misconduct in a course you cannot drop that course. DO NOT CHEAT and be proactive about avoiding the potential appearance of cheating.

Examples of Cheating Behavior (a non-exhaustive list)

Exams

- Learning what is on an exam from someone who has already taken it
- Using a false excuse to delay taking an exam
- Copying from another student on an exam **with or without** their knowledge
- Helping someone else cheat on an exam, i.e., seeing that they are copying your work and not reporting them.
- Using unauthorized information: Textbook, lecture notes, homework solutions, etc.
- Using unauthorized electronic devices to obtain information during an exam
- Using more time than allowed for an exam (take-home exam)
- Unauthorized communication during an exam
- Requesting a re-grade after modifying an exam

Homework

- Copying someone else's computer code when asked for individual work

- Working with others on an assignment when not authorized
- Receiving unpermitted help on an assignment
- Paraphrasing/copying material from a written/printed source without referencing it
- Paraphrasing/copying material from an internet source without referencing it
- Fabricating/falsifying a bibliography
- Turning in work **copied** from another
- Turning in work **done** by another
- Obtaining solutions from a database of previous years' assignments/exams

Other work

- Fabricating or falsifying lab data
- Fabricating or falsifying research data

Statement on Inclusivity (Sourced from ASEE website)

It is my intention that students from all backgrounds and perspectives will be well served by this course, and that the diversity that students bring to this class will be viewed as an asset. I welcome individuals of all ages, backgrounds, beliefs, ethnicities, genders, gender identities, gender expressions, national origins, religious affiliations or lack thereof, sexual orientations, socioeconomic background, family education level, ability – and other visible and nonvisible differences. All members of this class are expected to contribute to a respectful, welcoming and inclusive environment for every other member of the class. Your suggestions are encouraged and appreciated.

Other University Notices and Policies

Use of E-mail for Official Correspondence to Students

All students should become familiar with the University's official e-mail student notification policy. It is the student's responsibility to keep the University informed as to changes in his or her e-mail address. Students are expected to check e-mail on a frequent and regular basis in order to stay current with University-related communications, recognizing that certain communications may be time-critical. It is recommended that e-mail be checked daily, but at a minimum, twice per week. The complete text of this policy and instructions for updating your e-mail address are available at <http://www.utexas.edu/its/help/utmail/1564>.

Documented Disability Statement

Any student with a documented disability who requires academic accommodations should contact Services for Students with Disabilities (SSD) at (512) 471-6259 (voice) or 1-866-329-3986 (video phone). I will do my best to provide accommodations with an official accommodation letter from SSD. Please notify me as early in the semester as possible if you need disability-related accommodations. Given the size of the class, and my only support being a single TA, it will be up to students with SSD letters to arrange their own accommodations at the SSD office for extra time on midterms. Midterms at the SSD office may only be taken while the rest of the class is also sitting for the midterm. Unfortunately, no accommodations can be made for in-class quizzes. If you intend to record lectures, that is permitted. However, as a courtesy I ask that you inform me before you begin recording.

Behavior Concerns Advice Line (BCAL)

If you are worried about someone who is acting differently, you may use the Behavior Concerns Advice Line to discuss by phone your concerns about another individual's behavior. This service is provided through a partnership among the Office of the Dean of Students, the Counseling and Mental Health Center (CMHC), the Employee Assistance Program (EAP), and The University of Texas Police Department (UTPD). Call 512-232-5050 or visit <http://www.utexas.edu/safety/bcal>.

Sharing of Course Materials is Prohibited: No materials used in this class, including, but not limited to, lecture hand-outs, videos, assessments (quizzes, exams, papers, projects, homework assignments), in-class materials, review sheets, and additional problem sets, may be shared online or with anyone outside of the class unless you have my explicit, written permission. Unauthorized sharing of materials promotes cheating. It is a violation of the University's Student Honor Code and an act of academic dishonesty. I am well aware of the sites used for sharing materials, and any materials found online that are associated with you, or any suspected unauthorized sharing of materials, will be reported to Student Conduct and Academic Integrity in the Office of the Dean of Students. These reports can result in sanctions, including failure in the course.

Class Recordings: Class recordings are reserved only for students in this class for educational purposes and are protected under FERPA. The recordings should not be shared outside the class in any form. Violation of this restriction by a student could lead to Student Misconduct proceedings.

Handguns. Students in this class should be aware of the following university policies:

- Individuals who hold a license to carry are eligible to carry a concealed handgun on campus, including in most outdoor areas, buildings and spaces that are accessible to the public, and in classrooms.
- It is the responsibility of concealed-carry license holders to carry their handguns on or about their person at all times while on campus. Open carry is NOT permitted, meaning that a license holder may not carry a partially or wholly visible handgun on campus premises or on any university driveway, street, sidewalk or walkway, parking lot, parking garage, or other parking area.
- Students in this class who hold a license to carry are asked to review the university [policy](#) regarding campus carry.
- Legally, you can do many things, but does that mean you should?
- **Per my right, I prohibit carrying of handguns in my personal office unless I am also armed with a firearm.**

TITLE IX DISCLOSURE

Beginning January 1, 2020, Texas Education Code, Section 51.252 (formerly known as Senate Bill 212) requires all employees of Texas universities, including faculty, to report to the [Title IX Office](#) any information regarding incidents of sexual harassment, sexual assault, dating violence, or stalking that is disclosed to them. Texas law requires that all employees who witness or receive information about incidents of this type (including, but not limited to, written forms, applications, one-on-one conversations, class assignments, class discussions, or third-party reports) must report it to the Title IX Coordinator. Before talking with me, or with any faculty or staff member about a Title IX-related incident, please remember that I will be required to report this information.

Although graduate teaching and research assistants are not subject to Texas Education Code, Section 51.252, they are [mandatory reporters](#) under federal Title IX regulations and are required to report [a wide range of behaviors we refer to as sexual misconduct](#), including the types of misconduct covered under Texas Education Code, Section 51.252. Title IX of the Education Amendments of 1972 is a federal civil rights law that prohibits discrimination on the basis of sex – including pregnancy and parental status – in

educational programs and activities. The Title IX Office has developed supportive ways and compiled campus resources to support all impacted by a Title IX matter.

If you would like to speak with a case manager, who can provide support, resources, or academic accommodations, in the Title IX Office, please email: supportandresources@austin.utexas.edu. Case managers can also provide support, resources, and accommodations for pregnant, nursing, and parenting students.

For more information about reporting options and resources, please visit: <https://titleix.utexas.edu>, contact the Title IX Office via email at: titleix@austin.utexas.edu, or call 512-471-0419.